Traffic Datasets – Detailed Descriptions

1. SUMO Traffic Scenarios

Sumo scenarios dataset

SUMO (Simulation of Urban MObility) is an open-source traffic simulation platform widely used for traffic research.
SUMO scenarios are not a single dataset but collections of traffic simulations created using road networks, vehicle routes,
and traffic signal definitions.

Ex: TorontoSUMONetworks is an open-source platform designed for creating, manipulating, and simulating traffic networks in Toronto using SUMO. This tool provides a realistic simulation environment for transportation research, urban planning, and traffic management. Its flexibility makes it valuable for diverse research areas, including Intelligent Transportation Systems (ITS), Reinforcement Learning for Adaptive Traffic Signal Control (ATSC), and more. Toronto

Source of Data:
Most SUMO scenarios are generated from OpenStreetMap (OSM) road network data combined with synthetic or real traffic demand models.

Typical Files:
- network.net.xml : road network structure including intersections, lanes, and traffic lights
- routes.rou.xml : vehicle routes and departure times
- simulation.sumocfg : simulation configuration file
- additional.xml : optional traffic signal definitions and detectors

What the Dataset Contains:
- road network topology
- vehicle definitions
- routes and traffic demand
- traffic signal programs

Input to Algorithms:
- lane-level vehicle counts
- queue length
- vehicle positions

Output:
- travel time
- delay

- queue length
- throughput

Applications:
- traffic signal control
- route optimization
- congestion analysis
- autonomous vehicle testing

Related Paper:
Title: "Luxembourg SUMO Traffic (LuST) Scenario: 24 hours of mobility for vehicular networking research" ≈ 300+ citations
Link: https://ieeexplore.ieee.org/document/7385539

2. CityFlow Dataset

CityFlow
CityFlow is a large-scale multi-agent traffic simulation environment designed for reinforcement learning research.
Source of Data:
Road networks are derived from real cities or synthetically generated networks, combined with traffic demand generation.

Typical Files:
- roadnet.json : describes intersections, lanes, and traffic lights
- flow.json : defines vehicle flows, routes, and departure times

What the Dataset Contains:
- road network graph
- intersection structures
- vehicle flows
- traffic signal phases

Input to Algorithms:
- queue length per lane
- vehicle count
- waiting time

Output:
- average travel time
- average delay
- intersection throughput

Applications:
- multi-agent reinforcement learning

- adaptive traffic signal control
- large-scale traffic simulations

Related Paper:
Title: "CityFlow: A Multi-Agent Reinforcement Learning Environment for Large Scale City Traffic Scenario" ≈ **1200+ citations**
Link: https://arxiv.org/abs/1905.05217

3. PeMS Traffic Dataset

Data link
PeMS (Performance Measurement System) is a real-world traffic dataset collected from highway sensors in California.

Source of Data:
Loop detectors installed on highways by the California Department of Transportation (Caltrans).

What the Dataset Contains:
- traffic flow (vehicles per hour)
- vehicle speed
- lane occupancy
- timestamps

Dataset Size:
Hundreds of sensors and years of data depending on the region.

Input:
- sensor measurements every 5 minutes

Output:
- traffic forecasting targets such as future speed or flow

Applications:
- traffic flow prediction
- congestion forecasting
- spatio-temporal modeling

Related Paper:
Title: "Diffusion Convolutional Recurrent Neural Network: Data-Driven Traffic Forecasting"
Link: https://arxiv.org/abs/1707.01926 ≈ 1500+ citations

4. Manchester Traffic Dataset

https://roadtraffic.dft.gov.uk/downloads
The Manchester traffic dataset contains urban traffic data collected from road sensors and detectors in Manchester, UK.

Source of Data:
Traffic monitoring systems installed by local transportation authorities.

What the Dataset Contains:
- vehicle counts
- speed measurements
- timestamps
- sensor IDs

Applications:
- urban traffic modeling
- congestion analysis
- traffic prediction

5. Zenodo Traffic Flow Dataset for Traffic Signal Timing Optimization

Data link

This dataset was created specifically for studying Traffic Signal Timing Optimization (TSTO) using the SUMO simulator.

Source of Data:
Traffic volumes were generated and used to create vehicle routes using SUMO's JTRrouter tool.

Network Types Included:
Synthetic networks:
- Net-single1
- Net-single2
- Net-single3
- Net-double
- Net-2x2grid
- Net-3x3grid

Real-world networks:
- Cologne
- Ingolstadt

Dataset Size:
22 Traffic Signal Timing Adaptation Problem (TSAP) instances.
Each instance contains 1200 traffic demand samples.

What the Dataset Contains:
- traffic volume inputs (τ)
- route generation parameters

- SUMO network definitions
- vehicle departure times

Applications:
- traffic signal optimization
- reinforcement learning training
- traffic simulation experiments

Related Paper:
Title: "Traffic Signal Timing Optimization: From Evolution to Adaptation"
Link: https://zenodo.org/records/14653151

6. Hangzhou Traffic Signal Control Dataset

Data link

The Hangzhou traffic dataset is widely used for reinforcement learning research in traffic signal control.

Source of Data:
Traffic flow data from Hangzhou, China collected from cameras and taxi GPS trajectories.

Typical Files:
- roadnet.json : intersection and lane structure
- flow.json : vehicle flow definitions

Network Sizes:
- single intersection scenarios (1x1)
- medium networks (3x3 grid) 9 intersections
- larger networks (4x4 grid) 16 intersections

What the Dataset Contains:
- road network topology
- traffic signal phases
- vehicle routes and flows
- turning ratios derived from real traffic

Input:
- lane vehicle counts
- queue length
- waiting time

Output:
- average travel time - average delay

Applications:
- adaptive traffic signal control
- multi-agent reinforcement learning